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Adaptive Soundtracks in Video Games

Kirk Hogden, 001115381

Introduction to the Problem

- Music matters in a video game. *Plut, C. and Pasquier, P. (2019)*
- Linear soundtracks become tedious and eventually break immersion in player experiences. *Plut, C. and Pasquier, P. (2019)*
- Despite advantages of Adaptive music, games still choose to use linear soundtracks. *Plut, C. and Pasquier, P. (2019)*

Aim

Discovering how to remove repetitiveness in video game soundtracks, increasing immersion players feel while gaming.

Chosen Method

Generative

Main Theories

Using generative soundtracks can improve player immersion in video games, solving the issue of repetitiveness.

Much like the same way it's been proven sounds overall have an impact on player emotions. *Thiparpakul, P., Mokekhaow, S. and Supabanpot, K. (2021)*

Similar Products

Galactic Escape – A game made to test the tension that players experience during gameplay via music. *Plut, C. and Pasquier, P. (2019).*

How Can Video Game Atmosphere Affect Audience Emotion with Sound – A case study to see if video-game sounds effect player emotions during gameplay. *Thiparpakul, P., Mokekhaow, S. and Supabanpot, K. (2021).*

Proposed Solution

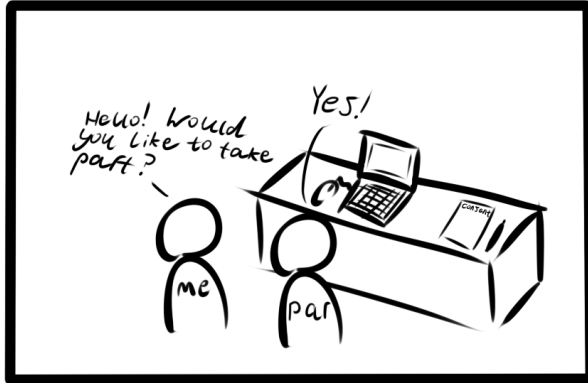
- A simple-to-play game of which uses different soundtrack methods.
- Participants are invited to play after they fill out a consent form.
- After playing for as long as they want, they're requested to partake in a survey which asks their thoughts relating to music.



Next slide shows storyboard for how this solution helps collect data.

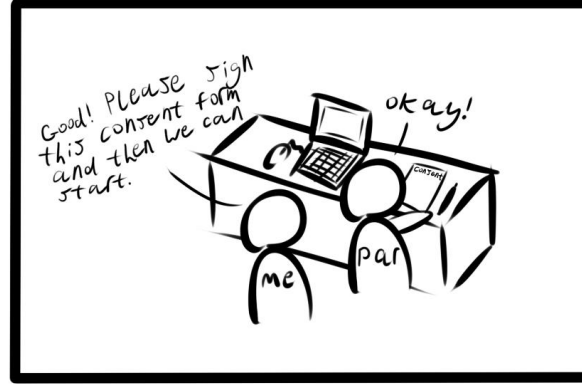
Storyboard

SLIDE ONE



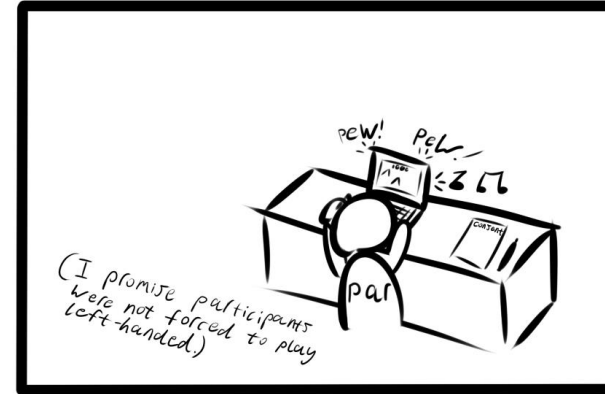
Participants are invited to partake in the user testing.

SLIDE TWO



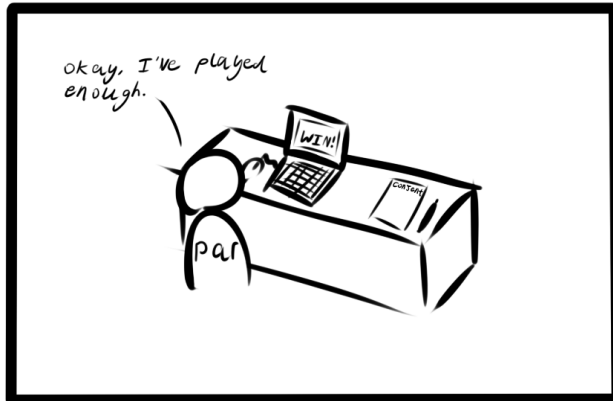
If they say yes, they're first asked to sign the consent form before they're allowed to officially take part.

SLIDE THREE



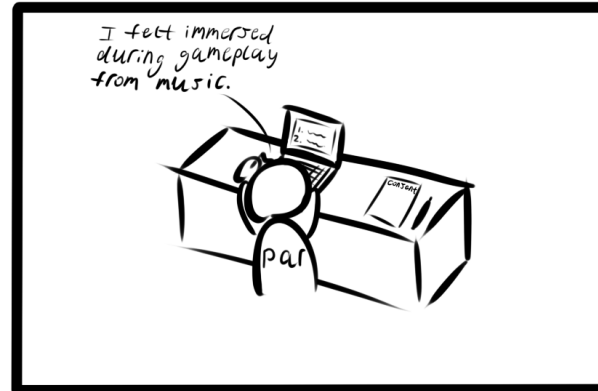
The participant is asked to play the game.

SLIDE FOUR



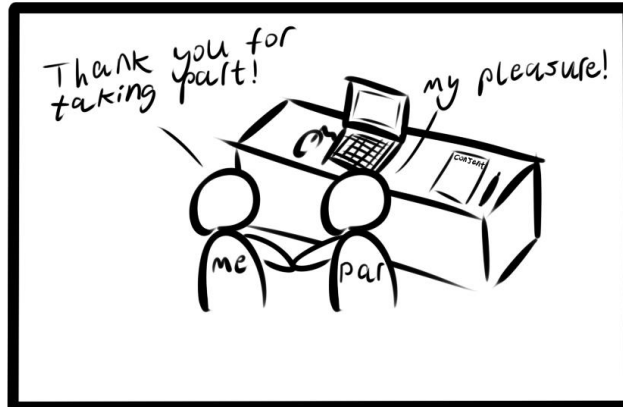
The participant can play for as long as they like. This allows them to take in the game.

SLIDE FIVE



Once the participant is finished playing, they're given a questionnaire regarding the game's music.

SLIDE SIX



The participant has finished their session, and their data is saved onto a cloud drive for research.

Prototyping – Written Idea

- The base prototype for the solution was done through a write-up.
- This was quick to create, and suitable as the game idea was kept small. (Adams and Joris Dormans, 2012, pp. 17-18)
- It narrows down the basic idea of gameplay.

Infinite Cosmos

Summary

The setting takes place in space, as hinted by the title. The player controls a space ship which is constantly moving forward. Enemy space ships are trying to pursue the player and eliminate them, using various attacks. As time goes on, the game gets progressively harder as tougher enemies approach. Can the player survive long enough and reach their destination safely?

Goal

The player must destroy enemy space ships that approach them, while trying to survive a time limit. When the time limit is reached, the player will see their final score given to them depending on how well they performed. The score can be increased by destroying or damaging enemies, as well as a bonus score that is rewarded just before the player sees their final amount. The bonus amount depends on how well the player has performed, such as how much health they had left over by the end of the session.

Player Mechanics

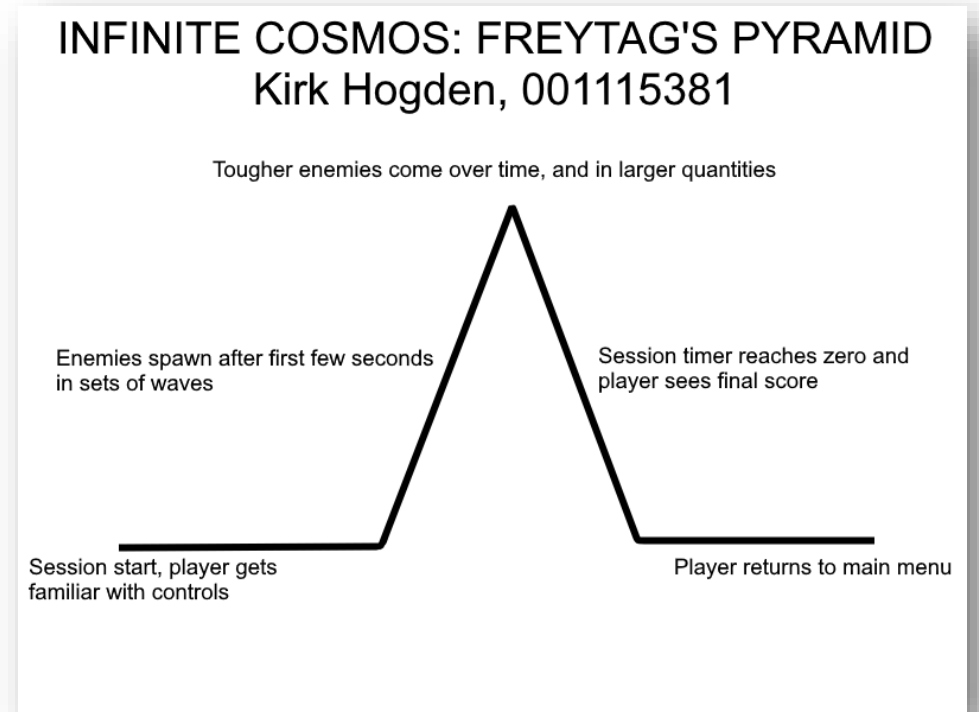
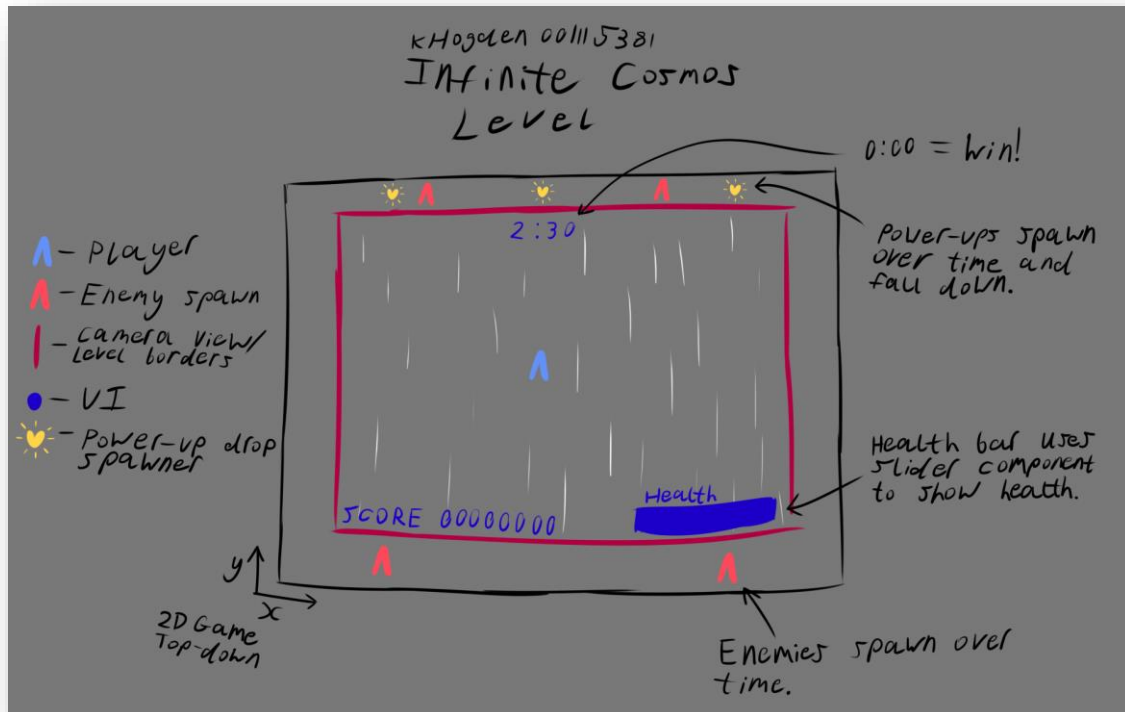
Move up, down, left, right with arrow keys. Left mouse fires bullets forwards. Bullets do little damage but have little cooldown time. Right click fires an explosive bomb backwards, does lots of damage but long cooldown. Player has one hundred percent health, which upon reaching zero percent results in game over.

Generative Music

The game will use generative adaptive music which uses parameters to calculate the player's situation. The music will sound calm when the player is performing well, however it will become more aggressive when the players is low on health.

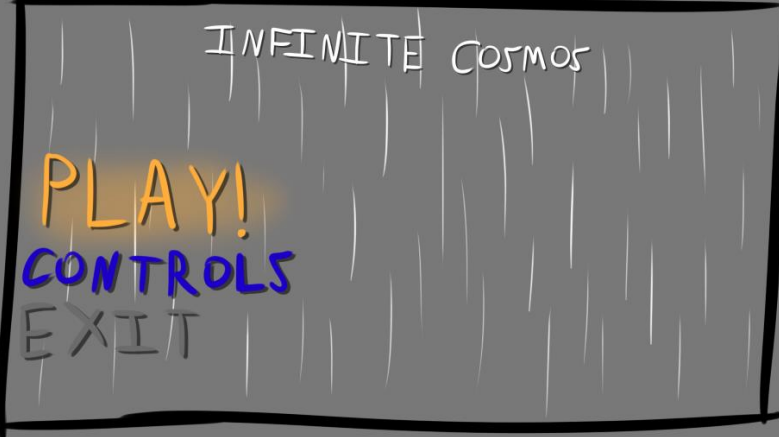
Prototyping – Designs

- A linear level was made so that each participant's gameplay experience goes the same order. (Kremer, 2009, p. 56)
- Pyramid is used to describe the UX journey that players go through when playing the game. (Kremer, 2009, pp. 67-79)



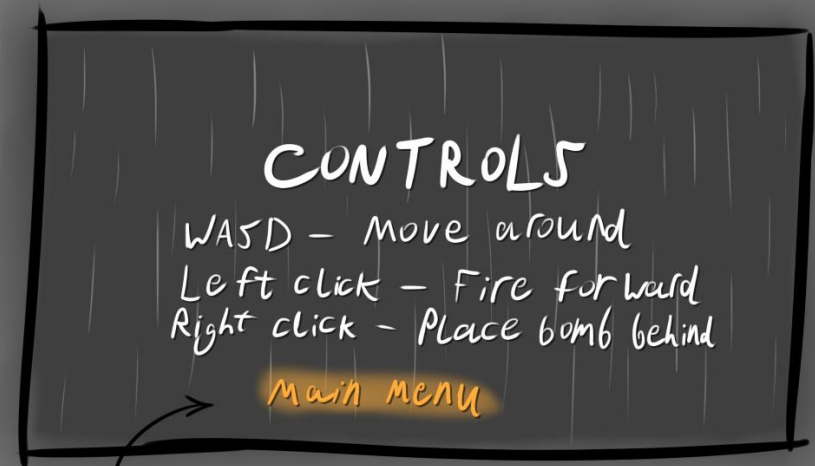
Prototyping – UI 1/3

Infinite Cosmos VI
Main Menu



- Player can execute a game session or see game controls.
- Glow indicates which button is highlighted.
- Generative music plays relaxing theme.

Infinite Cosmos VI
controls menu



- main menu button let's player return to main menu.

Prototyping – UI 2/3

Infinite Cosmos VI
Victory Menu

YOU WIN!



Bonus score: 5000
Score: 25000

- Shows when timer reaches zero.
- Player gains bonus score and sees final score.
- Generative music changes to a celebrative theme.
- Returns to main menu after short period.

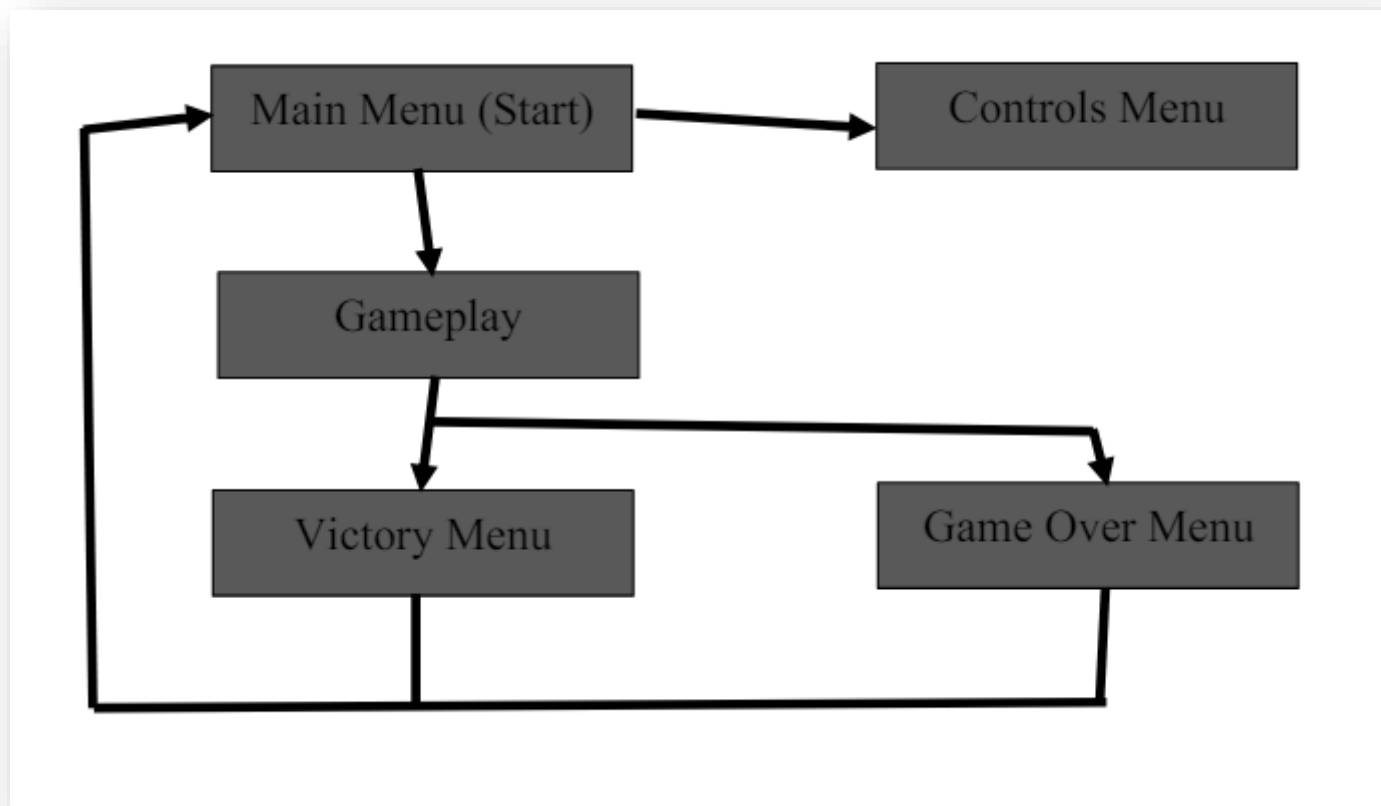
Infinite Cosmos VI
Game over UI

GAME OVER

Score: 00000000

- Shows when player dies.
- Player will get to see final score.
- Auto returns to main menu after short period.
- Generative music changes to be a sad theme.

Prototyping – UI 3/3



THE INFINITE COSMOS

This project uses CSound to play generative music, as a way to tackle the issue of repetitiveness in video games, and hopefully make players feel more immersed.

Demo Video

https://drive.google.com/file/d/1q_CWNehfZ6HrU2obfkbmDyLB1g_Qp5ca/view?usp=sharing

Exit

Developed by Kirk Hogden 001115381

Made with Unity, Programmed in C# with Microsoft Visual Studio 2022

Modelling in Blender

Music Composed by Richard Knight, and Steven Yi. Implemented via CSoundUnity package

Sound Effects composed in DrPetter's SFXR

Evaluation - Table

- Collected primary data was stored into tables.
- Answers were totalled to find the mean average of what most participants said.

'Q + i' presents questions, 'P + i' presents individual participants.

	Q1	Q2	Q3	Q4	Q5
P1	4	1	3	5	5
P2	4	1	5	5	4
P3	5	1	4	3	3
P4	4.5	2	1	5	5
P5	5	4	5	5	4
P6	5	1	5	5	5
P7	4	1	5	5	4
P8	5	1	4	4	4
P9	5	1	5	4	4
P10	2.5	4	2.5	2.5	1
P11	4	1	5	4	5
Mean	4.3	1.63...	4.04...	4.31...	4

Evaluation – Final Comments

Below is a list of commonly said things from participants during final comments.

- Feelings of better immersion and suspension during gameplay.
- It fit the aesthetic of the game, portraying 80's/90's arcade music.
- Others even felt opposite, that the game could've used war-like music and that it didn't fit with the intended narrative.
- Others couldn't hear the music too well due to sound effects being too loud.
(Useful feedback to bare in mind if this research project were to be repeated.)

Evaluation – Discussion

- Participants felt mostly happy overall about the music.
- This project aimed to increase immersion in music, and results show that was achieved.
- Despite the majority being positive results, some portrayed dissatisfaction.
- The reasonings behind most negative thoughts evolved around the way the music was implemented, rather than it being about generative music in general.
- It should be kept in mind these participants weren't aware they were listening to generative music, or at least not told directly.

Conclusion 1/2

- This project is considered a success!
- The results answered that generative music can be a potential method for video games, so long as the algorithm is well developed.
- They opposed what was initially stated by the composer of No Man's Sky who criticised generative music of being worse. (Pute & Pasquier, 2020)
- These answers could be used for future research projects relating to generative music.

Conclusion 2/2

- While this project is considered a success, psychological methods could've been a better way to collect data due to it's increased use in prior game music research projects by others. (Precht, Laney, Willis, & Samuels, 2014, pp. 3-4)
- It's imaginable the results could be different if the music was interactive through in-game parameters.
- In future research projects, features outside of the project's main focus should be carefully made. For example: The final comment stating about sound effects being too loud to even hear music.

References

Plut, C. and Pasquier, P. (2019). Generative Music in Video Games: State of the Art, Challenges, and Prospects. *Entertainment Computing*, 33, p.100337. doi:<https://doi.org/10.1016/j.entcom.2019.100337>.

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Prechtl, A., Laney, R., Willis, A. and Samuels, R. (2014). Methodological approaches to the evaluation of game music systems. In: *Proceedings of the 9th Audio Mostly: A Conference on Interaction With Sound*. New York : Association for Computing Machinery, pp.1–8.

Questions and Answers/Thank you (Last Slide)

The last few years being a student at University of Greenwich for this course have taught me beyond what I already learnt before with game development and design. I have greatly enjoyed being a student for this course, every tutor here has been amazing, and I am very thankful for the lessons taught. This has been a great final module to do before I reach the end of my school life.

Q/A - Do you have any questions?